

SignConnect : Real-Time Translation

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Problem Statement

Deaf/mute individuals often face challenges communicating with others who don't understand sign language, which can lead to isolation and difficulty accessing essential services.



Our Solution

An AR translator that allows a deaf/mute person to communicate via sign language, with their gestures translated into text, and a hearing person to respond via speech-to-text.



Setting

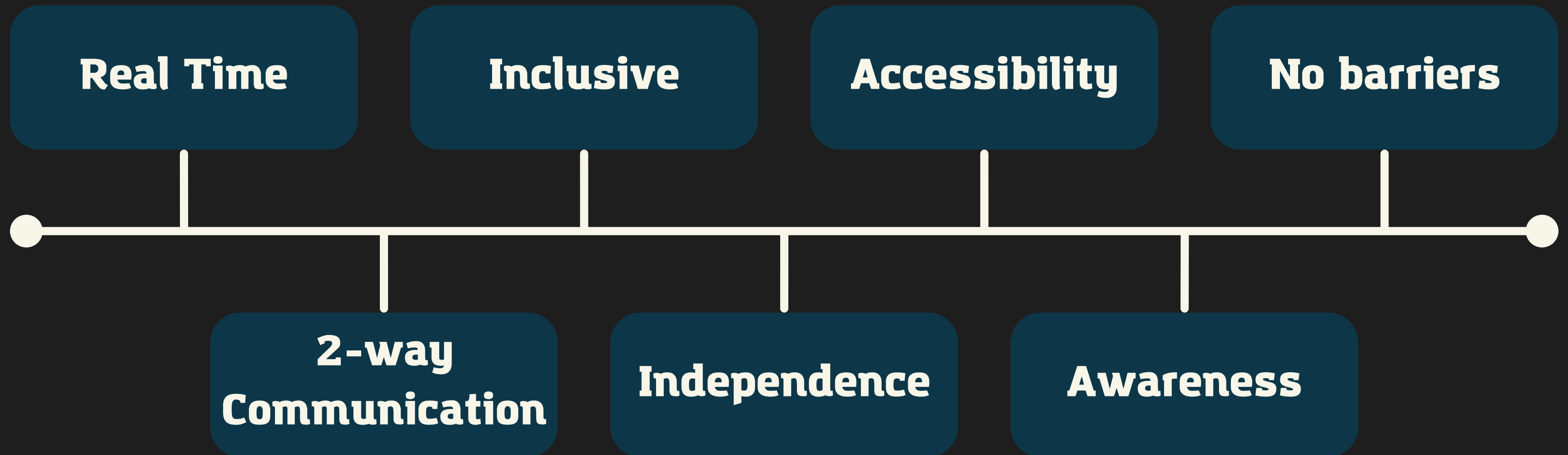
In locations such as hospitals, government offices, retail stores, and restaurants, communication can be challenging for deaf/mute individuals. This AR solution allows both parties to communicate naturally without needing a translator, fostering inclusivity and accessibility.

Impact

This technology can be beneficial in any setting that requires frequent interaction between deaf/mute and hearing individuals, improving accessibility and independence.



Purpose & Goals



Target Users

01

Deaf People

02

Mute People

03

People who dont know sign language



Features

- **Sign Language to Text:** Uses AR-based hand tracking to recognize gestures made by the deaf/mute individual and converts them into text for the hearing user.
- **Speech to Text:** Captures the hearing user's spoken words and displays them as text for the deaf/mute individual to read.
- **Real-Time Communication:** Both translations happen instantly, allowing for a natural, uninterrupted conversation.



Key Interaction

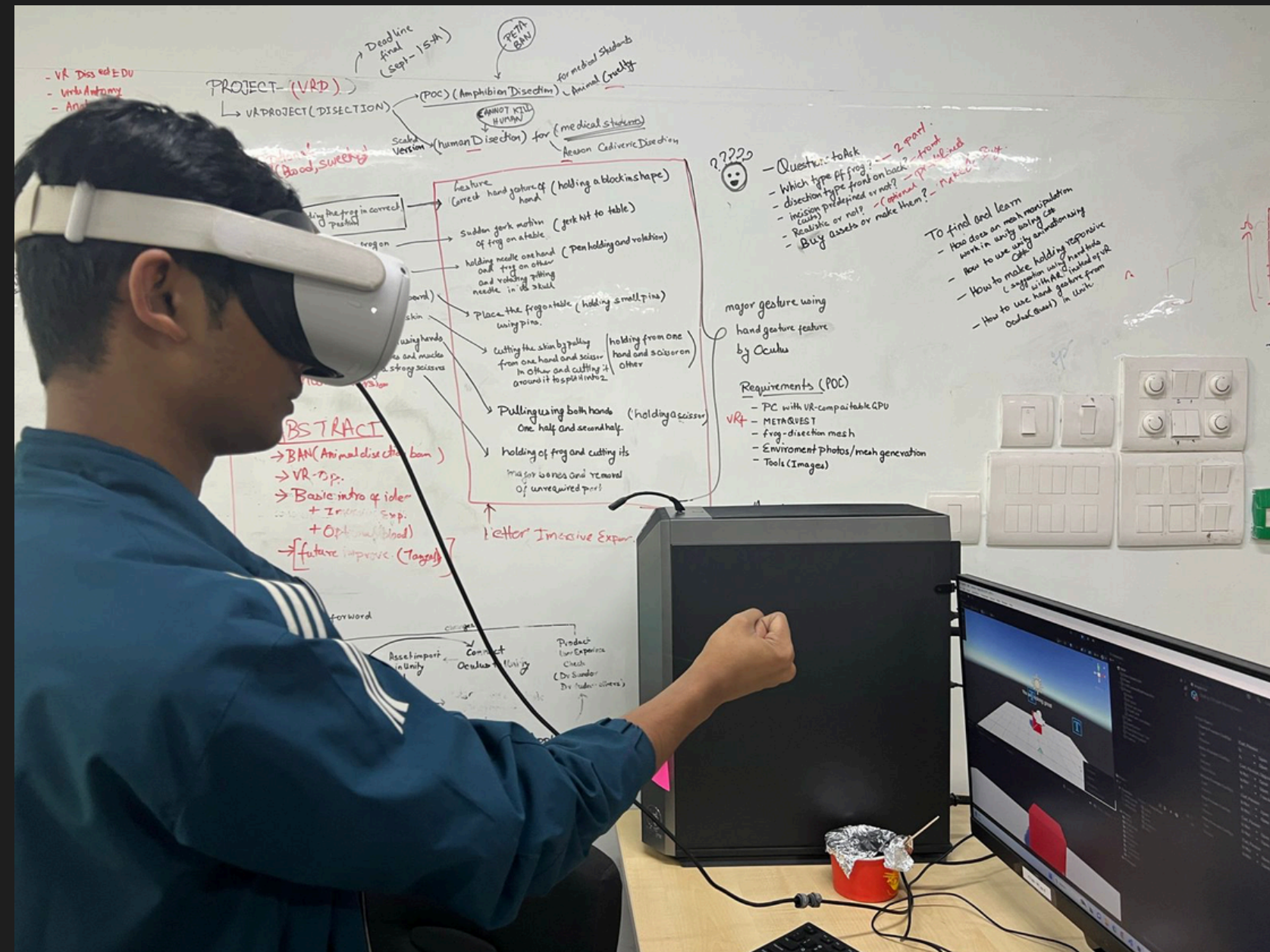
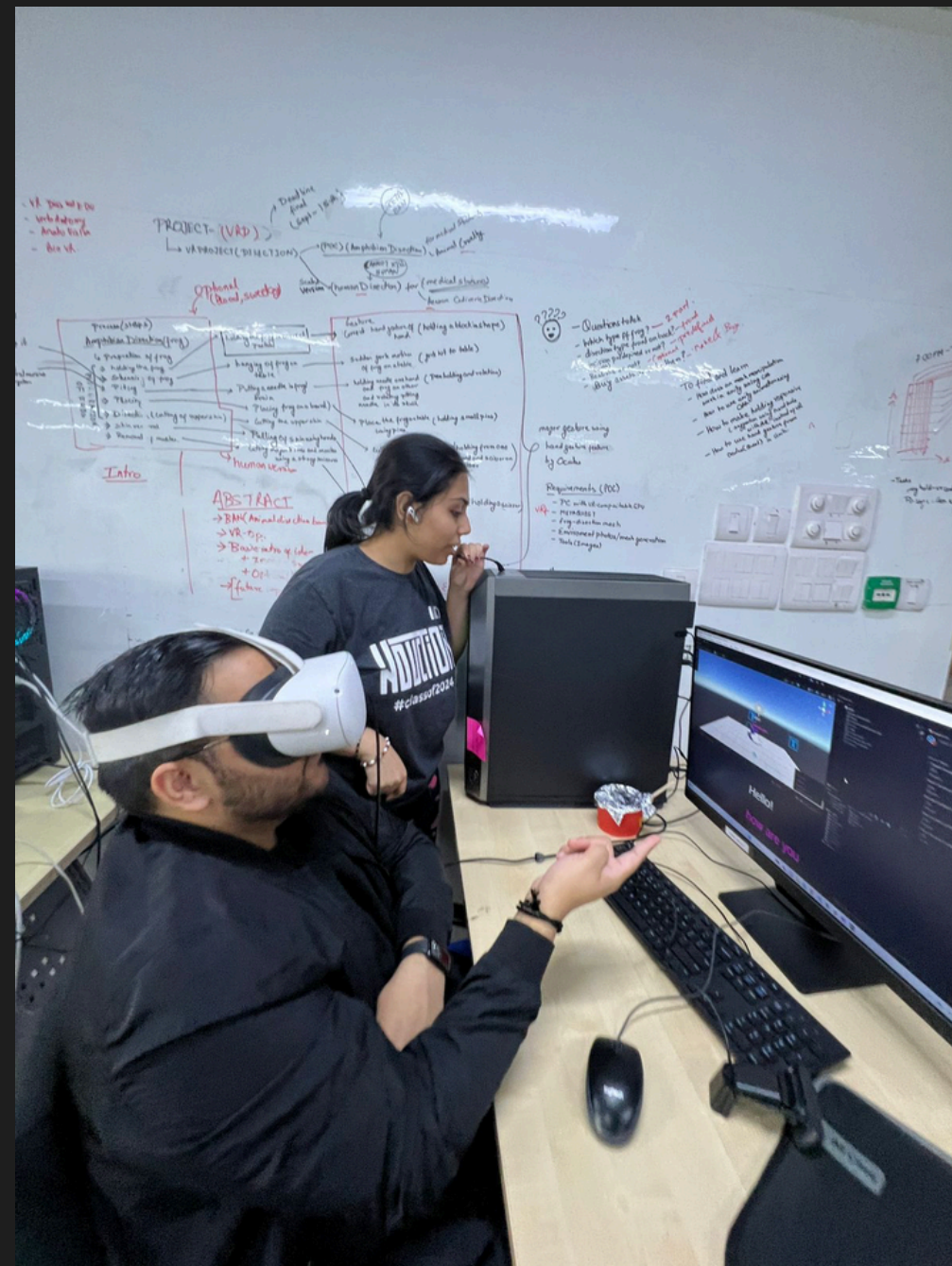
- **Sign Language User:** The deaf/mute individual signs, and hand tracking recognizes the gesture.
- **Text Display for Hearing User:** The app translates the gesture to text, showing it on-screen in real time for the hearing user to read.
- **Hearing User's Speech:** The hearing user speaks, and the app uses speech-to-text to display their words as text on the screen for the deaf/mute individual.



User Testing Participants



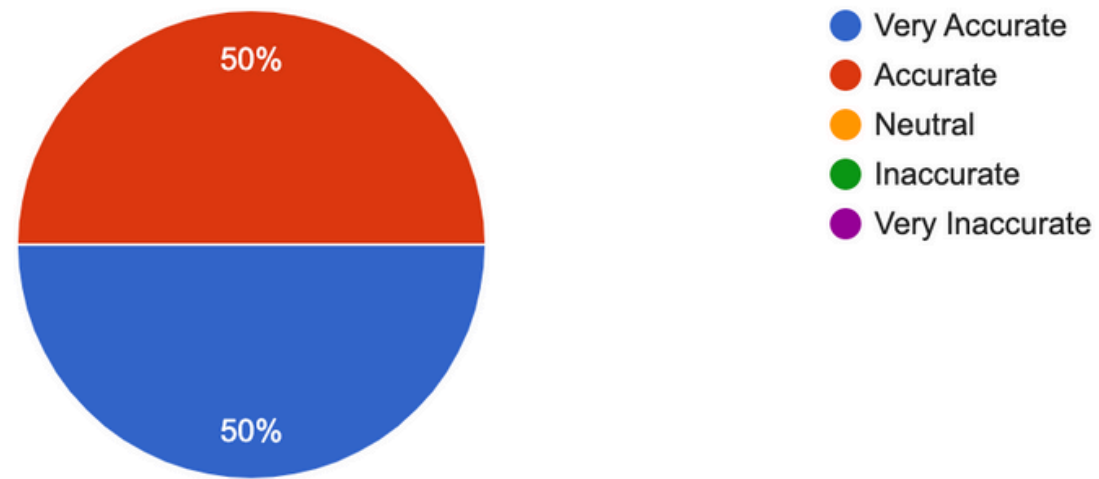
User Testing Participants



User Testing Feedback

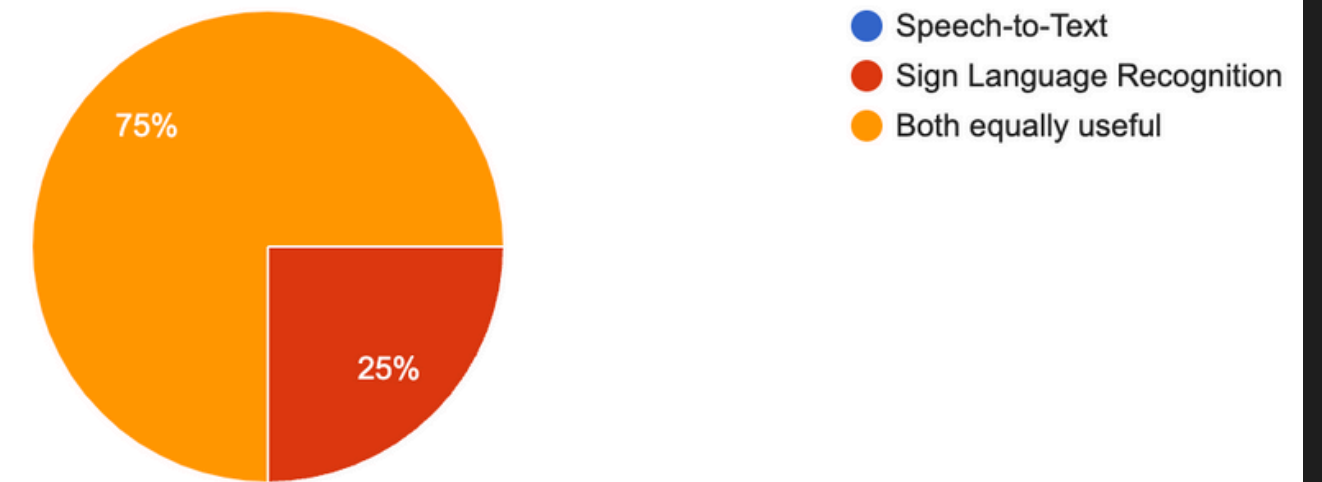
How accurate was the translation of **speech-to-text**?

4 responses



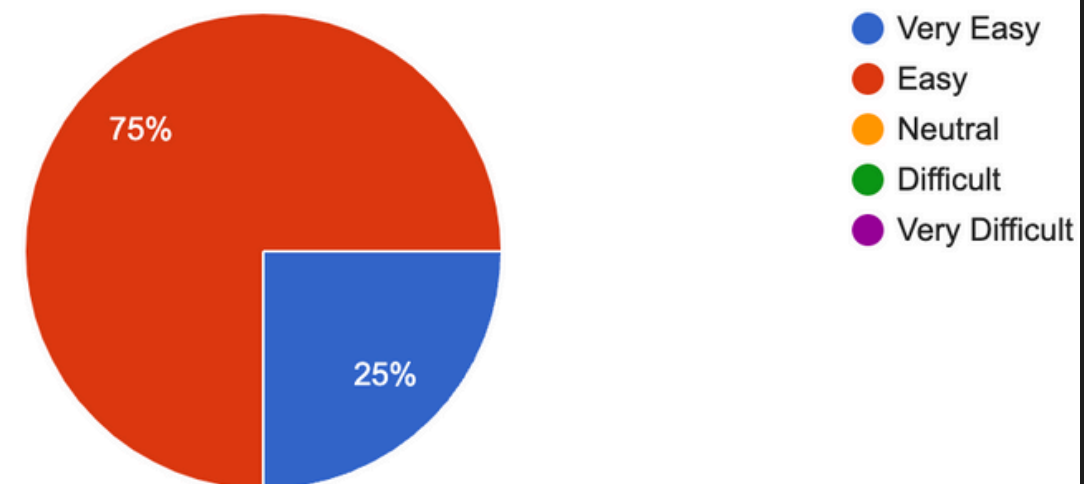
Which feature did you find most useful?

4 responses



How easy was it to use the app?

4 responses



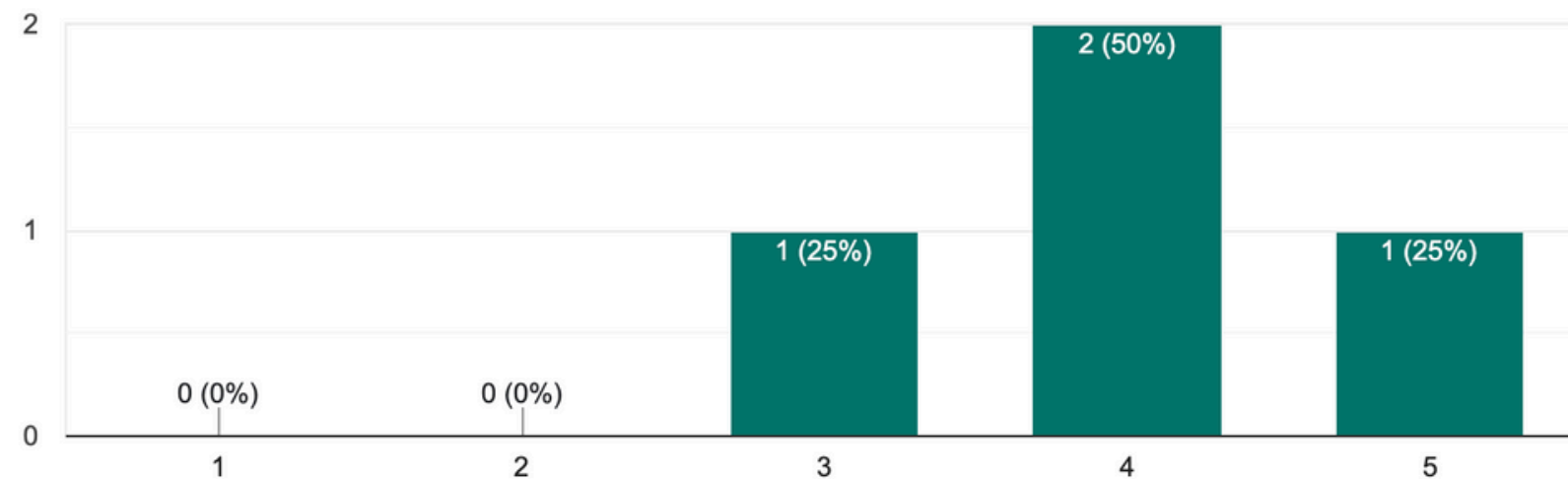
User Testing Feedback

This is giving us an idea of what all are the positives and the concerning drawbacks of the project which needs to be dealt with to enhance the user experience.

How satisfied are you with the app overall?

 Copy chart

4 responses



What changes or additional features would you like to see in this app?

2 responses

Add more signs

Adaptation to people of all heights

Research Paper

Our project, a sign language and speech-to-text converter, aligns well with the VAMR conference themes of enhancing user experience through mixed reality (MR) technologies. By addressing a significant accessibility challenge, this project aims to facilitate seamless communication between deaf/mute individuals and those who do not know sign language, utilizing MR to bridge gaps in understanding. It contributes to the conference's focus on interaction techniques and user-centered design, particularly in reducing cognitive load and enhancing inclusivity in mixed reality settings. This project highlights the potential of MR to deliver context-sensitive, accessible solutions in real-world applications and showcases the advancements in VAMR technologies that enhance everyday communication and accessibility.



Proposal Details

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Submission ID:	2674
Title:	SignConnect : Real Time Translation
Thematic Area:	Virtual, Augmented and Mixed Reality
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Demo Video



Demo Video



**Thank you for
listening!**